ExoHabitAI Project Documentation

Frontend

The frontend of the ExoHabitAI system provides a user-friendly interface that allows users to input planetary and stellar parameters and view habitability predictions.

Technologies Used

• HTML5 – Page structure
• CSS3 – Styling and layout
• Bootstrap – Responsive UI design
• JavaScript – Client-side logic and API communication

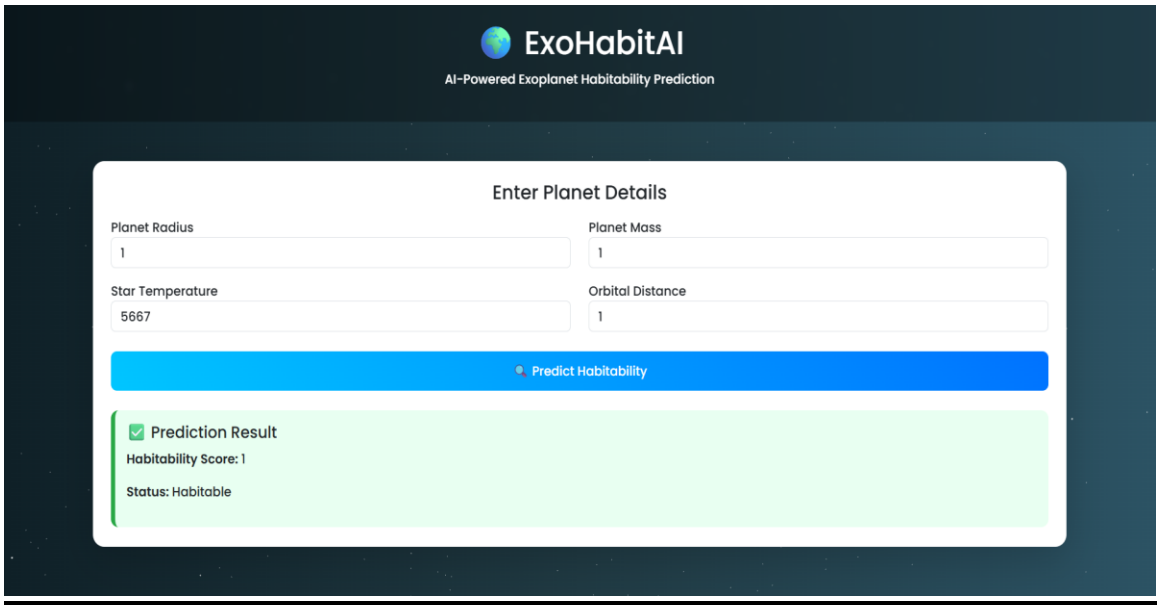
Frontend Features

• Input form for planet radius, mass, star temperature, and orbital distance
• Client-side validation for numeric values
• Fetch API used to send data to backend
• Dynamic display of habitability score and status
• Professional UI with gradient background and card layout

Frontend Workflow

User enters parameters → Frontend validates inputs → Data sent to backend → Backend returns prediction → Result displayed on UI

Output



Backend

The backend is built using Flask and integrates a trained machine learning model to predict the habitability of exoplanets based on input parameters.

Technologies Used

- Python
- Flask framework
- Scikit-learn (Machine Learning)
- Joblib/Pickle for model loading
- NumPy/Pandas for data handling

Backend Components

- Flask server to handle requests
- Model loading during server startup
- /predict API endpoint for receiving input
- Data processing and feature formatting
- Model prediction and probability scoring
- JSON response returned to frontend

Backend Workflow

Frontend sends JSON request → Flask receives request → Data processed → ML model predicts → JSON response returned

System Integration Flow

User → Frontend Form → Fetch API → Flask Backend → Machine Learning Model → Prediction Output → UI Display